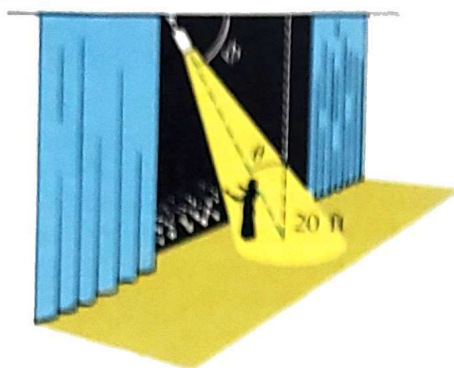


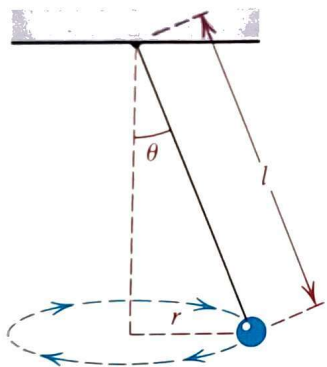
EXERCISE 48



- 49 A conical pendulum consists of a mass m , attached to a string of fixed length l , that travels around a circle of radius r at a fixed velocity v (see figure). As the velocity of the mass is increased, both the radius and the angle θ increase. Given that $r^2 = rg \tan \theta$, where g is a gravitational constant, find a relationship between the related rates

(a) dr/dt and $d\theta/dt$ (b) dv/dt and dr/dt

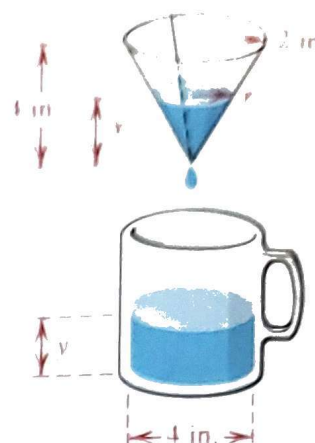
EXERCISE 49



- 50 Water in a paper conical filter drips into a cup as shown in the figure. Let x denote the height of the water in the filter and y the height of the water in the cup. If 10 in.³

of water is poured into the filter, find the relationship between dx/dt and dy/dt .

EXERCISE 50



- 51 Ship A is sailing north, and ship B is sailing west. Using an xy -plane, radar records the coordinates (in miles) of each ship at intervals of 1.25 minutes as shown in the following tables. Approximate the rate (in mi/hr) at which the distance between the ships is changing at $t = 5$.

t (min)	1.25	2.50	3.75	5.00
Ship A: x (mi)	1.77	1.77	1.77	1.77
y (mi)	2.71	3.03	3.35	3.67

t (min)	1.25	2.50	3.75	5.00
Ship B: x (mi)	5.24	5.52	5.80	6.08
y (mi)	1.24	1.24	1.24	1.24

- 52 Two variables x and y are functions of a variable t and are related by the formula
- $$1.31 \sin(2.56x) + \sqrt{y} = (x - 1)^2.$$
- If $dy/dt \approx 3.68$ when $x \approx 1.71$ and $y \approx 3.03$, approximate the corresponding value of dx/dt .

3.9 REVIEW EXERCISES

Exer. 1–2: Find $f'(x)$ directly from the definition of the derivative.

1 $f(x) = \frac{4}{3x^2 + 2}$

2 $f(x) = \sqrt{5 - 7x}$

Exer. 3–42: Find the first derivative.

3 $f(x) = 2x^3 - 7x + 2$

4 $k(x) = \frac{1}{x^4 - x^2 + 1}$

5 $g(t) = \sqrt{6t + 5}$

6 $h(t) = \frac{1}{\sqrt{6t + 5}}$

7 $F(z) = \sqrt[3]{7z^2 - 4z + 3}$

8 $f(w) = \sqrt[3]{3w^2}$

9 $G(x) = \frac{6}{(3x^2 - 1)^4}$

10 $H(x) = \frac{(3x^2 - 1)^4}{6}$

11 $F(r) = (r^2 - r^{-2})^{-2}$

12 $h(z) = [(z^2 - 1)^5 - 1]^5$

13 $g(x) = \sqrt[3]{(3x + 2)^4}$

14 $P(x) = (x + x^{-1})^2$

15 $r(s) = \left(\frac{8s^2 - 4}{1 - 9s^3} \right)^4$

16 $g(w) = \frac{(w - 1)(w - 3)}{(w + 1)(w + 3)}$

17 $F(x) = (x^6 + 1)^5(3x + 2)^3$

18 $k(z) = [z^2 + (z^2 + 9)^{1/2}]^{1/2}$

19 $k(s) = (2s^2 - 3s + 1)(9s - 1)^4$

$$20 \ p(x) = \frac{2x^4 + 3x^2 - 1}{x^2}$$

$$21 \ f(x) = 6x^2 - \frac{5}{x} + \frac{2}{\sqrt[3]{x^2}}$$

$$22 \ F(t) = \frac{5t^2 - 7}{t^2 + 2}$$

$$24 \ S(t) = \sqrt{t^2 + t + 1} \sqrt[3]{4t - 9}$$

$$25 \ g(r) = \sqrt{1 + \cos 2r}$$

$$27 \ f(x) = \sin^2(4x^3)$$

$$29 \ h(x) = (\sec x + \tan x)^5$$

$$31 \ f(x) = x^2 \cot 2x$$

$$33 \ K(\theta) = \frac{\sin 2\theta}{1 + \cos 2\theta}$$

$$35 \ g(x) = (\cos \sqrt[3]{x} - \sin \sqrt[3]{x})^3$$

$$36 \ f(x) = \frac{x}{2x + \sec^2 x}$$

$$38 \ k(\phi) = \frac{\sin \phi}{\cos \phi - \sin \phi}$$

$$39 \ F(x) = \sec 5x \tan 5x \sin 5x$$

$$40 \ H(z) = \sqrt{\sin \sqrt{z}}$$

$$42 \ f(x) = \csc^3 3x \cot^2 3x$$

Exer. 43–48: Assuming that the equation determines a differentiable function f such that $y = f(x)$, find y' .

$$43 \ 5x^3 - 2x^2y^2 + 4y^3 - 7 = 0$$

$$44 \ 3x^2 - xy^2 + y^{-1} = 1$$

$$45 \ \frac{\sqrt{x} + 1}{\sqrt{y} + 1} = y$$

$$47 \ xy^2 = \sin(x + 2y)$$

Exer. 49–50: Find equations of the tangent line and the normal line to the graph of f at P .

$$49 \ y = 2x - \frac{4}{\sqrt{x}}; \ P(4, 6)$$

$$50 \ x^2y - y^3 = 8; \ P(-3, 1)$$

51 Find the x -coordinates of all points on the graph of $y = 3x - \cos 2x$ at which the tangent line is perpendicular to the line $2x + 4y = 5$.

52 If $f(x) = \sin 2x - \cos 2x$ for $0 \leq x \leq 2\pi$, find the x -coordinates of all points on the graph of f at which the tangent line is horizontal.

Exer. 53–54: Find y' , y'' , and y''' .

$$53 \ y = 5x^3 + 4\sqrt{x}$$

$$54 \ y = 2x^2 - 3x - \cos 5x$$

$$23 \ f(w) = \sqrt{\frac{2w + 5}{7w - 9}}$$

$$26 \ g(z) = \csc\left(\frac{1}{z}\right) + \frac{1}{\sec z}$$

$$28 \ H(t) = (1 + \sin 3t)^3$$

$$30 \ K(r) = \sqrt[3]{r^3 + \csc 6r}$$

$$32 \ P(\theta) = \theta^2 \tan^2(\theta^2)$$

$$34 \ g(v) = \frac{1}{1 + \cos^2 2v}$$

$$37 \ G(u) = \frac{\csc u + 1}{\cot u + 1}$$

$$41 \ g(\theta) = \tan^4(\sqrt[4]{\theta})$$

55 If $x^2 + 4xy - y^2 = 8$, find y'' by implicit differentiation.

56 If $f(x) = x^3 - x^2 - 5x + 2$, find

(a) the x -coordinates of all points on the graph of f at which the tangent line is parallel to the line through $A(-3, 2)$ and $B(1, 14)$

(b) the value of f'' at each zero of f'

57 If $y = 3x^2 - 7$, find

(a) Δy (b) dy (c) $dy - \Delta y$

58 If $y = 5x/(x^2 + 1)$, find dy and use it to approximate the change in y if x changes from 2 to 1.98. What is the exact change in y ?

59 The side of an equilateral triangle is estimated to be 4 inches, with a maximum error of ± 0.03 inch. Use differentials to estimate the maximum error in the calculated area of the triangle. Approximate the percentage error.

60 If $s = 3r^2 - 2\sqrt{r+1}$ and $r = t^3 + t^2 + 1$, use the chain rule to find the value of ds/dt at $t = 1$.

61 If $f(x) = 2x^3 + x^2 - x + 1$ and $g(x) = x^5 + 4x^3 + 2x$, use differentials to approximate the change in $g(f(x))$ if x changes from -1 to -1.01 .

62 Use differentials to find a linear approximation of $\sqrt[3]{64.2}$.

63 Suppose f and g are functions such that $f(2) = -1$, $f'(2) = 4$, $f''(2) = -2$, $g(2) = -3$, $g'(2) = 2$, and $g''(2) = 1$. Find the value of each of the following at $x = 2$.

$$(a) (2f - 3g)' \quad (b) (2f - 3g)'' \quad (c) (fg)'$$

$$(d) (fg)'' \quad (e) \left(\frac{f}{g}\right)' \quad (f) \left(\frac{f}{g}\right)''$$

64 Refer to Exercise 87 in Section 3.6. Let f be an odd function and g an even function such that $f(3) = -3$, $f'(3) = 7$, $g(3) = -3$, and $g'(3) = -5$. Find $(f \circ g)'(3)$ and $(g \circ f)'(3)$.

65 Determine where the graph of f has a vertical tangent line or a cusp.

$$(a) f(x) = 3(x + 1)^{1/3} - 4 \quad (b) f(x) = 2(x - 8)^{2/3} - 1$$

$$66 \text{ Let } f(x) = \begin{cases} (2x - 1)^3 & \text{if } x \geq 2 \\ 5x^2 + 34x - 61 & \text{if } x < 2 \end{cases}$$

Determine if f is differentiable at 2.

67 The Stefan-Boltzmann law states that the radiant energy emitted from a unit area of a black surface is given by $R = kT^4$, where R is the rate of emission per unit area, T is the temperature (in $^\circ\text{K}$), and k is a constant. If the error in the measurement of T is 0.5%, find the resulting percentage error in the calculated value of R .

68 Let V and S denote the volume and surface area, respectively, of a spherical balloon. If the diameter is 8 centimeters and the volume increases by 12 cm^3 , use differentials to approximate the change in S .